

3ème rapport - Projet Image - Single Image Haze Removal Using Dark Channel Prior

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Work carried out during the session

We tried to drastically speed up the dark channel computation as using native Python *for loops* made it extruciatingly slow. We tried by using both Opencv's *erode* function that does the same thing as our loops and obtained the exact same result.

Similarly, we tried to parallelise the *for loop* version using *Taichi* (python parallelisation library) which got us the same speed and the exact same result

Example of an estimated dark channel yielded by our now faster code:



Figure 1: Dark Channel Computation Result

Difficulties encountered

Rediscovering the taichi library can be extrenuous since it almost changes the language's paradigm.

Future sessions

Continue pipeline implementations and test atmospheric light results.